

Report 25

19 -acquiring memory(47_48_49)-17_golden

Title:- Operation BlueBalance

Subtitle:- APT28 Emulation & Incident Investigation

Prepared By:

Suad Abbas Hussein

Bootcamp:

Purple Team / Golden

Supervisor:

Mohammed Baqer Hasan

Anmar Mohammed

Date Assigned: 2026/6/4

Date Submitted: 2026/6/22

Table of Contents

Executive Summary	2
Scenario Description	3
Part A – Setup	3
Step 1 – Create Evidence Directory.....	4
Step 2 – Download NulltackKatz.py.....	5
Step 3 – Prepare Mimikatz.....	5
Step 4 – Configure Gmail for Exfiltration	6
Step 5 – Start Monitoring Stack	6
Part B – Execute Attack	7
Task 1 – Run NulltackKatz	7
Part C: Detection & Investigation	10
Task 2 – Email Analysis (T1566.001 Detection)	10
Task 3 – Sysmon Analysis (T1003.001 Detection)	12
Task 4 – Process Tree Analysis.....	15
Task 5 — Network Traffic Analysis (T1041 Detection).....	17
Task 6 – Memory Forensics (Volatility 3).....	20
MITRE ATT&CK Mapping	23
Detection Rule Writing	23
Sysmon Detection Rule Checklist	24
Task 7 APT28 Full Threat Intelligence Study	25
Part 1 – Group Identity & Attribution	25
Part 2 – Historical Operations	26
Part 3: Technical TTP Analysis.....	26
Part 4: Infrastructure & IoCs.....	27
Part 5: Defensive Recommendations	28
Knowledge Graph (KG)	29
Conclusion	30

Introduction

This purple team scenario is designed to simulate the Tactics, Techniques, and Procedures (TTPs) used by APT28, one of the most active and dangerous cyber espionage groups operating today. Through hands-on emulation and detection exercises, you will step into the role of both attacker and defender, experiencing firsthand how this Russian GRU-affiliated threat group operates in the real world.

Your mission is to detect, analyze, and respond to a simulated APT28-style credential theft attack using industry-standard blue team tools including Sysmon, Wireshark, DumpIt, and Volatility 3. By the end of this scenario, you will understand how APT28 moves from an innocent-looking email to stolen credentials, and most importantly, how to stop them.

Executive Summary

This report describes the investigation of a simulated phishing attack as part of the APT28 emulation exercise.

The scenario started when a user received a phishing email that appeared to be a security compliance notification. The email instructed the user to download and run a file called NulltackKatz.py . After the file was executed, a report containing credential-related information was generated and sent by email.

To investigate the incident, a memory dump was collected using DumpIt and analyzed using Volatility 3. During the analysis, several processes, network connections, and credential-related artifacts were identified.

The purpose of this investigation was to understand how the attack was carried out, collect evidence from the affected system, and map the observed activity to the MITRE ATT&CK framework.

During the investigation, I confirmed that the phishing simulation successfully led to credential dumping activity and the exfiltration report was sent through Gmail services.. Evidence was collected from Sysmon logs, Wireshark captures, and memory analysis using Volatility 3.

Scenario Description

The Attack

On a routine Tuesday morning, a senior analyst at the National Finance Authority receives an urgent email from "IT Security." The email contains an attachment named NulltackKatz.py and threatens system lockout if not installed immediately.

The analyst, concerned about salary processing, downloads and runs the attachment. Unknown to them, the tool performs:

- Discovery — Maps local users, network configuration, and running processes
- Privilege Escalation — Enables SeDebugPrivilege for LSASS access
- Credential Dumping — Executes Mimikatz to extract passwords and NTLM hashes
- Exfiltration — Sends stolen credentials to an external Gmail account

Part A – Setup

Before starting the APT28 simulation, all required tools were prepared and checked. Sysmon was installed for event monitoring, Wireshark was prepared for network traffic capture, DumpIt and Volatility 3 were prepared for memory analysis, and both Mimikatz and NulltackKatz were downloaded as required by the scenario.

A Gmail App Password was also created and configured so that NulltackKatz could send the simulated exfiltration report to the configured email account during the exercise.

Required Tools Preparation

The following tools were used throughout this APT28 Purple Team simulation:

Tool	Purpose	Where to Get
Sysmon	Endpoint monitoring	Microsoft Sysinternals
Wireshark	Network traffic capture	wireshark.org
DumpIt	Memory acquisition	Magnet Forensics
Volatility 3	Memory forensics	https://github.com/AnmaR3B/volatility3
Mimikatz	Credential dumping (simulated)	https://github.com/AnmaR3B/mimikatz
NulltackKatz.py	APT28 emulation tool	https://drive.google.com/file/d/1tpFjOmPWBGw21YLneFRgqN6st8lehZd5/view?usp=drive_link

Step 1 – Create Evidence Directory

The required evidence folders were created to store logs and investigation artifacts generated during the scenario.

Command Used:

```
mkdir C:\NulltackLab  
mkdir C:\NulltackLab\logs  
mkdir C:\NulltackLab\evidence
```

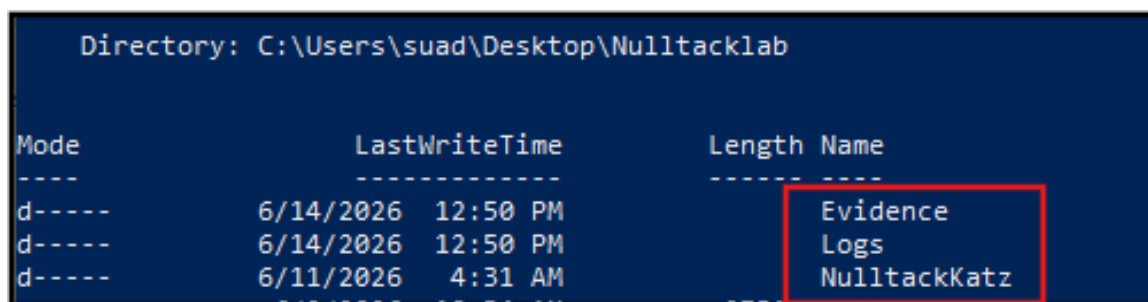
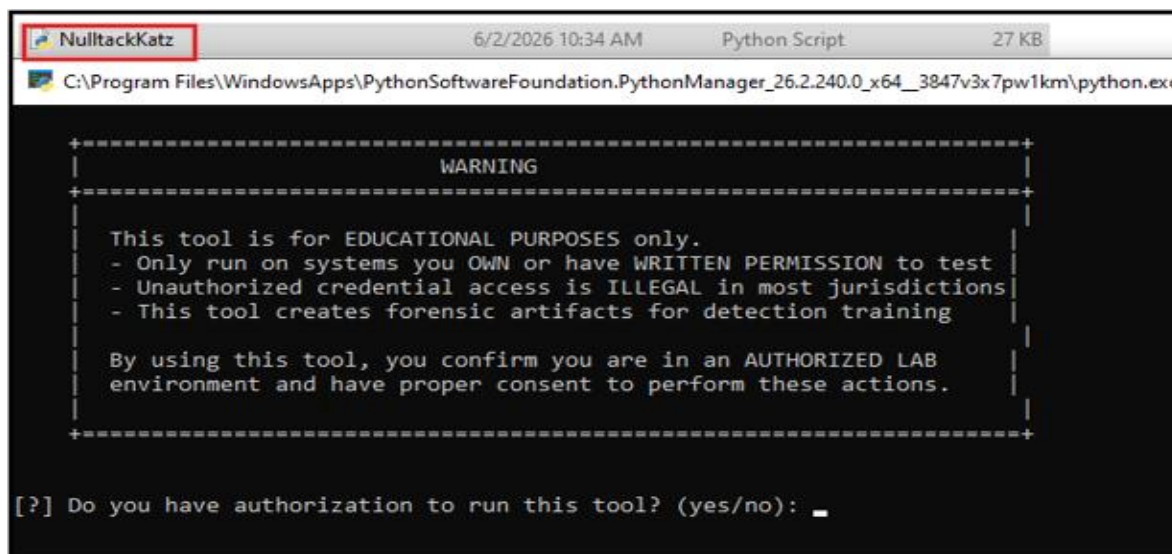


Figure 1 : Evidence Directory Structure

The directories were successfully created and were used to store logs, evidence files, and investigation artifacts generated during the APT28 simulation.

Step 2 – Download NulltackKatz.py

The NulltackKatz tool was downloaded and prepared for use during the APT28 simulation.



```
NulltackKatz 6/2/2026 10:34 AM Python Script 27 KB
C:\Program Files\WindowsApps\PythonSoftwareFoundation.PythonManager_26.2.240.0_x64_3847v3x7pw1km\python.exe

+-----+
| WARNING |
+-----+

This tool is for EDUCATIONAL PURPOSES only.
- Only run on systems you OWN or have WRITTEN PERMISSION to test
- Unauthorized credential access is ILLEGAL in most jurisdictions
- This tool creates forensic artifacts for detection training

By using this tool, you confirm you are in an AUTHORIZED LAB
environment and have proper consent to perform these actions.

[?] Do you have authorization to run this tool? (yes/no): _
```

Figure 2 : NulltackKatz

Step 3 – Prepare Mimikatz

Mimikatz was downloaded and prepared as one of the required tools for the scenario. The tool was successfully opened and verified before starting the simulation.



```
mimikatz 6/13/2026 4:21 PM Application 1,221 KB
mimikatz 2.2.0 x64 (oe.eo)

.#####. mimikatz 2.2.0 (x64) #18362 Feb 29 2020 11:13:36
.## ^ ##. "A La Vie, A L'Amour" - (oe.eo)
## / \ ## /*** Benjamin DELPY `gentilkiwi` ( benjamin@gentilkiwi.com )
## \ / ## > http://blog.gentilkiwi.com/mimikatz
'## v #' Vincent LE TOUX ( vincent.letoux@gmail.com )
'#####' > http://pingcastle.com / http://mysmartlogon.com ***/

mimikatz #
```

Figure 3 : Mimikatz Verification

Step 4 – Configure Gmail for Exfiltration

A Gmail App Password was created and configured to allow the exfiltration report generated during the simulation to be delivered to the configured email account.

Step 5 – Start Monitoring Stack

Before running the simulation, Sysmon and Wireshark were prepared to monitor system and network activity during the exercise.

Command Used:

```
Get-Service Sysmon
```

The result confirmed that the Sysmon service was running successfully.

```
PS C:\Users\suad\Desktop\Nulltacklab> Get-Service Sysmon

Status      Name      DisplayName
-----
Running Sysmon Sysmon
```

Figure 4 : Sysmon Service Verification

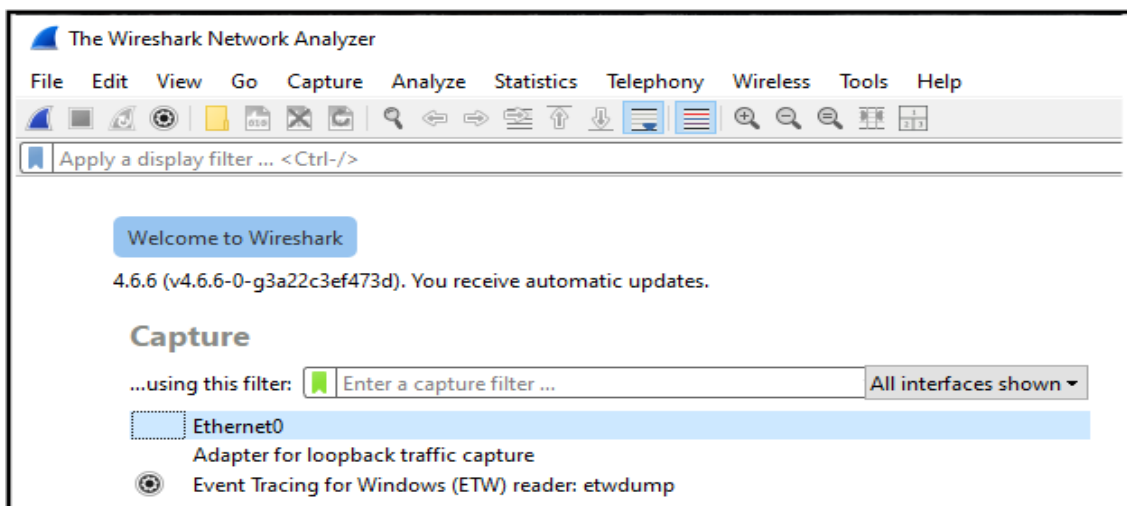
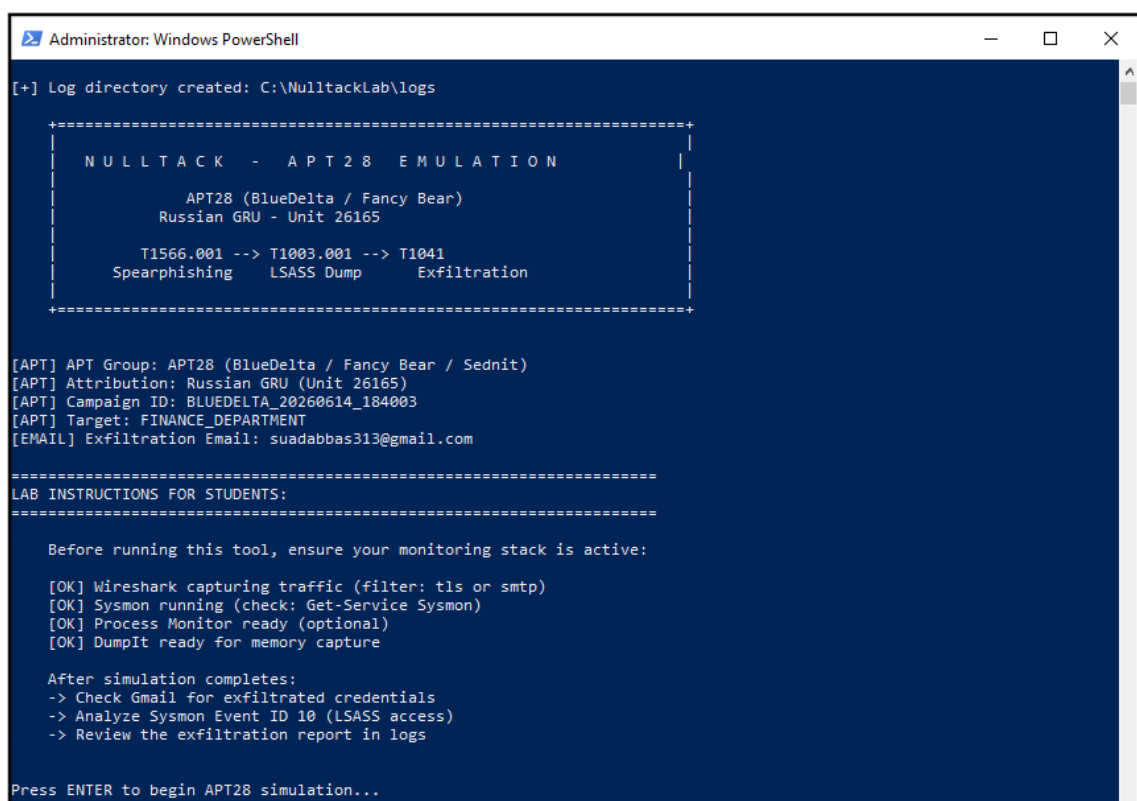


Figure 5 : Wireshark Ready for Capture

Part B – Execute Attack

Task 1 – Run NulltackKatz

The NulltackKatz tool was run using PowerShell as Administrator. During the simulation, the tool performed system discovery, simulated credential dumping, and generated an exfiltration report that was sent to the configured Gmail account. The simulation completed successfully, and the collected information could be reviewed through the received email report.



```
Administrator: Windows PowerShell

[+] Log directory created: C:\NulltackLab\logs

+-----+
| NULLTACK - APT28 EMULATION |
+-----+
|
|   APT28 (BlueDelta / Fancy Bear)
|   Russian GRU - Unit 26165
|
|   T1566.001 --> T1003.001 --> T1041
|   Spearphishing   LSASS Dump   Exfiltration
|
+-----+

[APT] APT Group: APT28 (BlueDelta / Fancy Bear / Sednit)
[APT] Attribution: Russian GRU (Unit 26165)
[APT] Campaign ID: BLUEDELTA_20260614_184003
[APT] Target: FINANCE_DEPARTMENT
[EMAIL] Exfiltration Email: suadabbas313@gmail.com

=====
LAB INSTRUCTIONS FOR STUDENTS:
=====

Before running this tool, ensure your monitoring stack is active:

[OK] Wireshark capturing traffic (filter: tls or smtp)
[OK] Sysmon running (check: Get-Service Sysmon)
[OK] Process Monitor ready (optional)
[OK] DumpIt ready for memory capture

After simulation completes:
-> Check Gmail for exfiltrated credentials
-> Analyze Sysmon Event ID 10 (LSASS access)
-> Review the exfiltration report in logs

Press ENTER to begin APT28 simulation...
```

Figure 6 : NulltackKatz Execution

After the simulation completed successfully, an exfiltration report was generated and sent to the configured Gmail account. This confirmed that the simulated attack was able to collect information and deliver the report through email.


```

Administrator: Windows PowerShell

Report sent to: suadabbas313@gmail.com
Subject: [APT28-BLUEDELTA-BLUEDELTA_20260614_184003] Compromised: DESKTOP-QQDS8EG
Local copy saved to: C:\NulltackLab\logs\exfiltration_report_BLUEDELTA_20260614_184003.txt

=====
APT28 SIMULATION COMPLETE
=====

[+] Duration: 3.64 seconds
[+] Campaign ID: BLUEDELTA_20260614_184003
[+] Logs saved to: C:\NulltackLab\logs

=====
MITRE ATT&CK TECHNIQUES DEMONSTRATED
=====

+-----+
| Technique | Description |
+-----+
| T1566.001 | Spearphishing Attachment - Email with payload |
| T1204.002 | Malicious File Execution - User runs tool |
| T1087.001 | Local Account Discovery - 'net user' command |
| T1016 | Network Config Discovery - 'ipconfig /all' |
| T1057 | Process Discovery - 'tasklist' for AV/EDR |
| T1134 | Token Manipulation - SeDebugPrivilege |
| T1003.001 | LSASS Credential Dumping - Mimikatz |
| T1041 | Exfiltration over C2 Channel - Gmail SMTP |
+-----+

[+] Evidence files for SOC investigation:
[FILE] C:\NulltackLab\logs\phishing_email_BLUEDELTA_20260614_184003.txt
[FILE] C:\NulltackLab\logs\mimikatz_output_BLUEDELTA_20260614_184003.txt
[FILE] C:\NulltackLab\logs\exfiltration_report_BLUEDELTA_20260614_184003.txt

[EMAIL] Check your Gmail: suadabbas313@gmail.com
Look for subject: [APT28-BLUEDELTA-...] Compromised: ...

=====
NEXT STEPS FOR SOC TEAM:
=====

1. Analyze Sysmon Event ID 10 for LSASS handle access (GrantedAccess=0x1010)
2. Run Volatility on memory dump: windows.handles | grep lsass
3. Review Wireshark capture for TLS to Gmail IPs
4. Document IOCs in incident report
5. Create detection rules for this TTP

[Press Enter to exit...]
    
```

Figure 7: NulltackKatz Simulation Completed

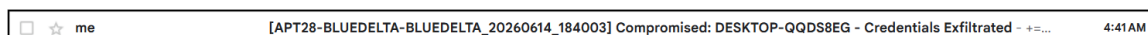


Figure 8 : Exfiltration email received after executing the attack simulation

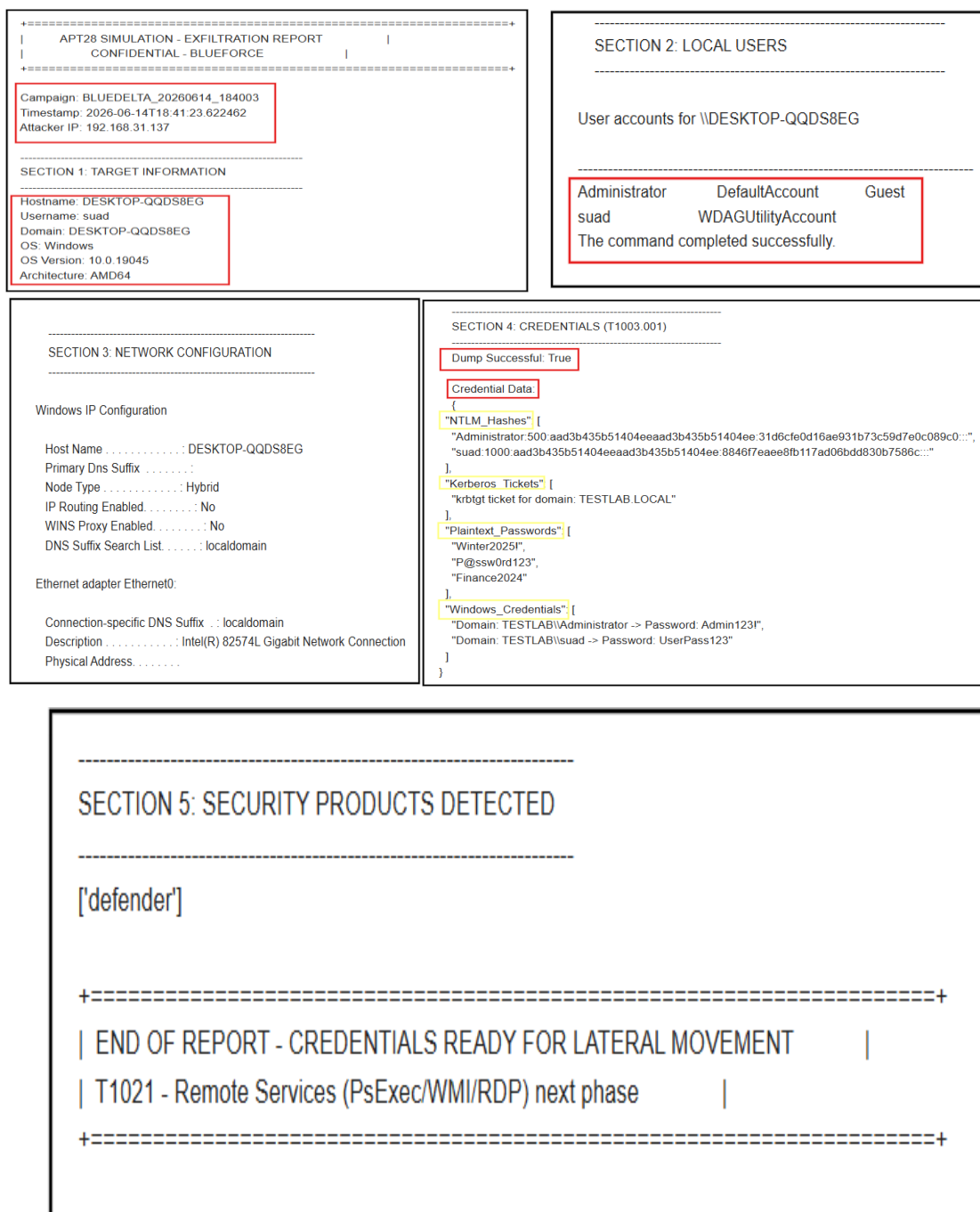


Figure 9 : Contents of the exfiltration report generated by NulltackKatz.

Part C: Detection & Investigation

Task 2 – Email Analysis (T1566.001 Detection)

Email Investigation

The investigation started by reviewing the suspicious email received by the user. The email claimed to be related to a Department of Homeland Security directive and requested the user to download and execute a file called NulltackKatz.

The sender displayed the name Anmar Al_Falahi, while the message content claimed to be from Elena Vasquez, IT Security Compliance Lead. This mismatch between the sender and the claimed identity is a suspicious indicator.

The email also created a sense of urgency by stating that failure to complete the requested action before the deadline would result in network access suspension and compliance escalation. Such pressure tactics are commonly used in phishing campaigns.

In addition, the email instructed the recipient to download and execute a file from a Google Drive link. Requesting users to download and run a file from an external source is another strong phishing indicator.

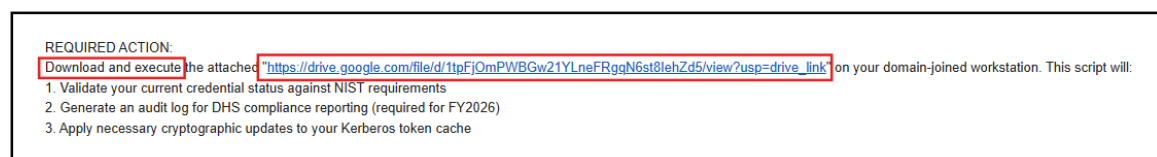


Figure 10 : The phishing email instructed the victim to download and execute a file from an external Google Drive link.

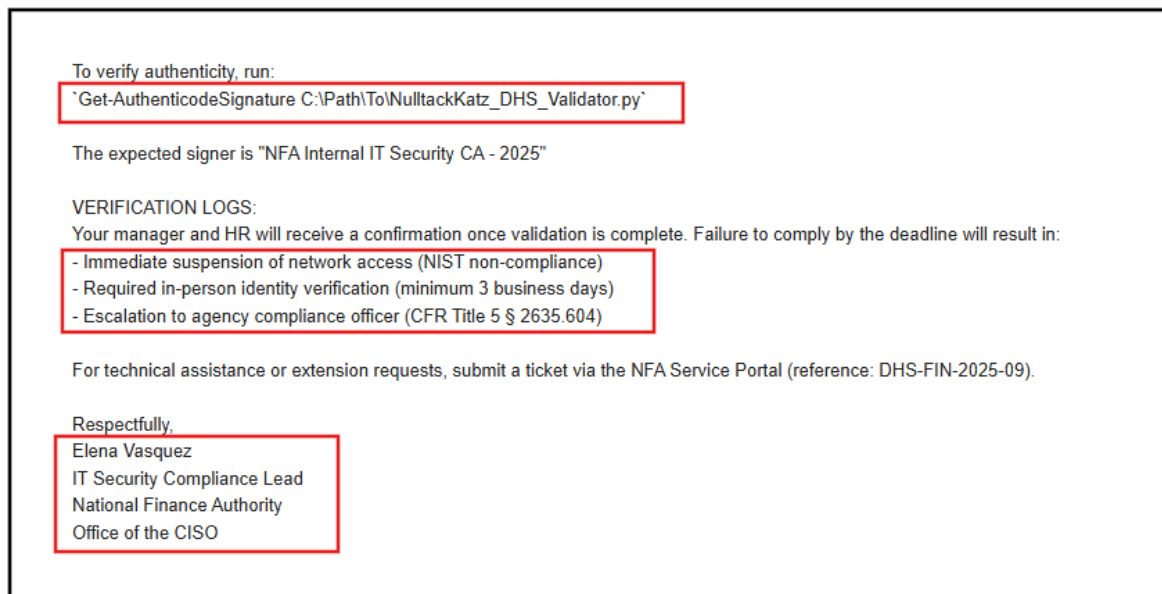


Figure 11 : The phishing email used urgency and fear tactics by threatening network access suspension and compliance escalation if the requested action was not completed.

Generated Phishing Email Analysis

The phishing email generated by NulltackKatz was reviewed to identify phishing indicators and answer the investigation questions.

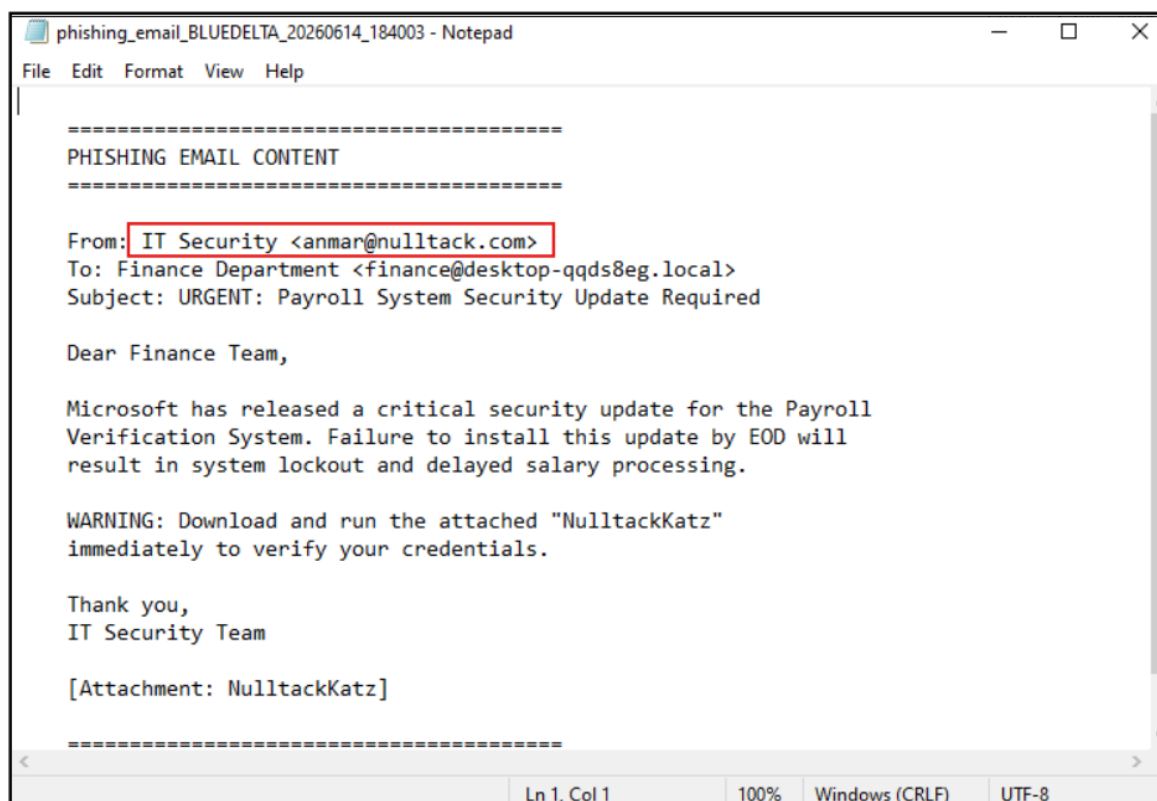


Figure 12 : Generated Phishing Email (NulltackKatz)

Task 3 – Sysmon Analysis (T1003.001 Detection)

Investigation Process

As part of the investigation, I reviewed the Sysmon logs to look for Event ID 10 entries related to LSASS process access. Event ID 10 is commonly used to detect processes attempting to access LSASS memory, which may indicate credential dumping activity.

Command used:

```
Get-WinEvent -FilterHashtable @{  
  LogName='Microsoft-Windows-Sysmon/Operational'  
  ID=10  
} -MaxEvents 20 | Format-List
```

```
PS C:\Users\suaad\Downloads> Get-WinEvent -FilterHashtable @{
>> LogName='Microsoft-Windows-Sysmon/Operational'
>> ID=10
>> } -MaxEvents 20 | Format-List
Get-WinEvent : No events were found that match the specified selection criteria.
At line:1 char:1
+ Get-WinEvent -FilterHashtable @{
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (:) [Get-WinEvent], Exception
+ FullyQualifiedErrorId : NoMatchingEventsFound,Microsoft.PowerShell.Commands.GetWinEventCommand
```

Figure 13 : Event ID 10 Query Result (No Events Found)

The query did not return any matching events, and Sysmon reported that no Event ID 10 entries were found in the log.

Event ID 10 was not generated during this exercise. However, Sysmon captured other relevant events, including process creation activity related to Mimikatz, which helped support the investigation and timeline analysis.

To verify that Sysmon was configured correctly, I reviewed the active Sysmon configuration file. The review confirmed that Process Access monitoring was enabled, indicating that the absence of Event ID 10 was related to the scenario activity rather than a configuration issue.

```
PS C:\Users\suaad\Downloads> sysmon -c
- ProcessAccess onmatch: include combine rules using 'Or'
```

Figure 14 : Sysmon Configuration Showing ProcessAccess Monitoring Enabled

As part of the laboratory exercise, I executed Mimikatz to generate activity that could potentially be detected by Sysmon.

```
PS C:\Users\suaad\Downloads> .\mimikatz.exe

.#####.   mimikatz 2.2.0 (x64) #18362 Feb 29 2020 11:13:36
.## ^ ##.   "A La Vie, A L'Amour" - (oe.eo)
## / \ ##   /** Benjamin DELPY `gentilkiwi` ( benjamin@gentilkiwi.com )
## \ / ##   > http://blog.gentilkiwi.com/mimikatz
'## v #'    Vincent LE TOUX ( vincent.letoux@gmail.com )
'#####'    > http://pingcastle.com / http://mysmartlogon.com   ***/

mimikatz # privilege::debug
Privilege '20' OK
```

Figure 15 : Mimikatz Execution

```
mimikatz # sekurlsa::logonPasswords

Authentication Id : 0 ; 512730 (00000000:0007d2da)
Session          : Interactive from 1
User Name        : suad
Domain           : DESKTOP-QQDS8EG
Logon Server     : DESKTOP-QQDS8EG
Logon Time       : 6/14/2026 6:30:20 PM
SID              : S-1-5-21-3825611489-1665044178-313730354-1002

msv :
  [00000003] Primary
  * Username : suad
  * Domain   : DESKTOP-QQDS8EG
  * NTLM     : ada02c39cba9e011575ed8d7b26dab6c
  * SHA1     : d5b625ed491f42d5624ea8cd64e760448d2d55c7
  * DPAPI    : d5b625ed491f42d5624ea8cd64e76044

tspkg :
wdigest :
  * Username : suad
  * Domain   : DESKTOP-QQDS8EG
  * Password : (null)
kerberos :
  * Username : suad
  * Domain   : DESKTOP-QQDS8EG
  * Password : (null)
ssp : KO
credman :
```

Figure 16 : Mimikatz Output

After executing Mimikatz, I reviewed the Sysmon logs again. Sysmon successfully recorded process creation activity related to the tool, which confirmed that monitoring was working correctly.

However, no Event ID 10 entries related to LSASS access were found during the investigation. Although ProcessAccess monitoring was enabled, no LSASS Process Access event was generated during this laboratory exercise. Therefore, the findings in this section are based on the evidence that was collected from the system during the investigation.

Question	Answer
What is the SourceImage that accessed lsass.exe?	No Event ID 10 was found during the investigation.
What is the GrantedAccess value?	Not available because Event ID 10 was not generated.
What does 0x1010 represent?	Not observed during this investigation.
What is the TargetImage?	Not available because Event ID 10 was not generated.
Why is this event the PRIMARY IoC for T1003.001?	Event ID 10 is commonly used to identify attempts to access LSASS memory, which is a common technique associated with credential dumping (T1003.001). However, no Event ID 10 was observed during this investigation.

Task 4 – Process Tree Analysis

Investigation Process

During the investigation, I reviewed Sysmon Event ID 1 logs to identify processes related to Mimikatz and NulltackKatz activity.

The logs showed that **mimikatz.exe** was executed from the Downloads directory. I also reviewed the process creation details to understand how the tool was launched and which user account was associated with the activity.

The Sysmon event showed that the parent process was **Windows PowerShell (powershell.exe)**, indicating that PowerShell was used to launch the executable.

The command line recorded in the event was:

"C:\Users\sud\Downloads\mimikatz.exe"

The event also showed that the process was executed under the following user account:

DESKTOP-QQDS8EG\sud

By reviewing the process creation event, I was able to identify the parent-child process relationship and determine the user context associated with the activity.

Command used:

```
Get-WinEvent -FilterHashtable

@{
  LogName='Microsoft-Windows-Sysmon/Operational'
  ID=1
} |

Where-Object{
  $_.Message -like "mimikatz" -or $_.Message -like
  "NulltackKatz"
} | Format-List
```

```
PS C:\Users\suaud> Get-WinEvent -FilterHashtable @{
>   LogName='Microsoft-Windows-Sysmon/Operational'
>   ID=1
> } | Where-Object {
>   $_.Message -like "mimikatz" -or $_.Message -like "NulltackKatz"
> } | Format-List

TimeCreated      : 6/15/2026 3:49:35 PM
ProviderName     : Microsoft-Windows-Sysmon
Id              : 1
Message         : Process Create:
                  RuleName: -
                  UtcTime: 2026-06-15 22:49:35.802
                  ProcessGuid: {d8da4647-817f-6a30-4703-000000002b00}
                  ProcessId: 5012
                  Image: C:\Users\suaud\Downloads\mimikatz.exe
                  FileVersion: 2.2.0.0
                  Description: mimikatz for Windows
                  Product: mimikatz
                  Company: gentilkiwi (Benjamin DELPY)
                  OriginalFileName: mimikatz.exe
                  CommandLine: "C:\Users\suaud\Downloads\mimikatz.exe"
                  CurrentDirectory: C:\Users\suaud\Downloads\
                  User: DESKTOP-QQDS8EG\suaud
                  LogonGuid: {d8da4647-55ac-6a2f-afd2-070000000000}
                  TerminalSessionId: 1
                  IntegrityLevel: High
                  Hashes: MD5=E930805FE23891D198C354A4209BE3E, SHA256=92804FAA82175DC501D73E814663058C78C0A042675A89372663578CFB96C50, IMPHASH=1355327F6CA3430B3D0BE6E0ACDA71EA
                  ParentProcessGuid: {d8da4647-65c2-6a30-c102-000000002b00}
                  ParentProcessId: 8816
                  ParentImage: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
                  ParentCommandLine: "C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe"
                  ParentUser: DESKTOP-QQDS8EG\suaud
```

Figure 17 : Sysmon Event ID 1 showing Mimikatz execution details, including the parent process, command line, and user context.

Question	Answer
What was the parent process of NulltackKatz?	Windows PowerShell (powershell.exe)
What command-line arguments were passed?	"C:\Users\suaud\Downloads\ mimikatz.exe "
What user context did it run under?	DESKTOP-QQDS8EG\suaud

Task 5 — Network Traffic Analysis (T1041 Detection)

Network Investigation

I started by opening the Wireshark capture file and applying the following filter:

tls and ip.dst

This filter helped me focus on outbound encrypted connections initiated by the host system.

After applying the filter, I observed multiple TLSv1.2 connections between the host machine (192.168.31.137) and several external IP addresses. During the review, I identified TLS handshake traffic including Client Hello, Server Hello, and encrypted Application Data packets. This indicated that encrypted communication was taking place between the host and external systems.

No.	Time	Source	Destination	Protocol	Length	Info
9	7.345166200	192.168.31.137	135.232.92.137	TLSv1.2	265	Client Hello (SNI=slscr.update.microsoft.com)
15	7.533362900	192.168.31.137	135.232.92.137	TLSv1.2	212	Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
18	7.712240700	192.168.31.137	135.232.92.137	TLSv1.2	141	Application Data
20	7.725648000	192.168.31.137	135.232.92.137	TLSv1.2	323	Application Data
23	7.888449600	192.168.31.137	135.232.92.137	TLSv1.2	92	Application Data
2330	125.561302100	192.168.31.137	142.251.127.108	TLSv1.3	1575	Client Hello (SNI=smtp.gmail.com)
2339	125.854973300	192.168.31.137	142.251.127.108	TLSv1.3	164	Change Cipher Spec, Application Data, Application Data
2341	125.855390500	192.168.31.137	142.251.127.108	TLSv1.3	110	Application Data
2344	126.039564400	192.168.31.137	142.251.127.108	TLSv1.3	149	Application Data
2347	126.443396600	192.168.31.137	142.251.127.108	TLSv1.3	82	Application Data
2370	131.129692200	192.168.31.137	142.251.127.108	TLSv1.3	1575	Client Hello (SNI=smtp.gmail.com)
2377	131.301459700	192.168.31.137	142.251.127.108	TLSv1.3	164	Change Cipher Spec, Application Data, Application Data
2379	131.301959400	192.168.31.137	142.251.127.108	TLSv1.3	110	Application Data
2383	131.473838100	192.168.31.137	142.251.127.108	TLSv1.3	149	Application Data
2387	131.751470200	192.168.31.137	142.251.127.108	TLSv1.3	122	Application Data
2390	131.837679400	192.168.31.137	142.251.127.108	TLSv1.3	110	Application Data
2393	131.923861500	192.168.31.137	142.251.127.108	TLSv1.3	82	Application Data
2398	132.941177800	192.168.31.137	142.251.127.108	TLSv1.3	3825	Application Data
2405	133.776223200	192.168.31.137	142.251.127.108	TLSv1.3	82	Application Data
15016	690.289112000	192.168.31.137	150.171.22.17	TLSv1.2	1849	Client Hello (SNI=config.edge.skype.com)
15030	690.508224900	192.168.31.137	150.171.22.17	TLSv1.2	212	Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
15039	690.612819900	192.168.31.137	150.171.22.17	TLSv1.2	843	Application Data
2538	192.315385600	192.168.31.137	150.171.27.11	TLSv1.2	1878	Client Hello (SNI=edge.microsoft.com)
2557	192.576588100	192.168.31.137	150.171.27.11	TLSv1.2	212	Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
2562	192.703585200	192.168.31.137	150.171.27.11	TLSv1.2	1368	Application Data
2563	192.703811000	192.168.31.137	150.171.27.11	TLSv1.2	5770	Application Data
2449	189.164533700	192.168.31.137	150.171.28.12	TLSv1.2	250	Client Hello (SNI=api.msn.com)
2459	189.398701900	192.168.31.137	150.171.28.12	TLSv1.2	212	Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
2461	189.403483500	192.168.31.137	150.171.28.12	TLSv1.2	4129	Application Data
15274	851.466946800	192.168.31.137	150.171.28.12	TLSv1.2	250	Client Hello (SNI=api.msn.com)
15283	851.607713800	192.168.31.137	150.171.28.12	TLSv1.2	212	Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
15285	851.610685200	192.168.31.137	150.171.28.12	TLSv1.2	4129	Application Data

Figure 18 : During the review, I identified a TLS connection to smtp.gmail.com. The traffic showed the TLS handshake process followed by encrypted Application Data packets, indicating that data was exchanged through an encrypted connection.

Several external IP addresses were observed during the capture. Some destinations exchanged more data than others and were selected for further review as part of the investigation.

Ethernet - 1	IPv4 - 33	IPv6	TCP - 49	UDP - 2										
Address A	Address B	Packets	Bytes	Stream ID	Total Packets	Percent Filtered	Packets A - B	Bytes A - B	Packets B - A	Bytes B - A	Rel Start	Duration	Bits/s A - B	Bits/s B - A
192.168.31.137	2.16.158.155	155	135 kb	50	698	22.21%	57	14 kb	98	121 kb	1151.032450000	47.626882	2305 bits/s	20 kbps
192.168.31.137	132.196.74.210	42	104 kb	2	266	15.79%	22	83 kb	20	21 kb	8.483205300	19.153906	34 kbps	8978 bits/s
192.168.31.137	150.171.28.12	55	73 kb	23	177	31.07%	9	14 kb	46	59 kb	189.055736600	961.294146	114 bits/s	490 bits/s
192.168.31.137	2.22.114.43	27	20 kb	13	76	35.53%	9	3 kb	18	18 kb	28.068036200	740.111771	30 bits/s	190 bits/s
192.168.31.137	51.105.71.137	26	18 kb	35	65	40.00%	14	14 kb	12	4 kb	32.1783803600	29.13212	3834 bits/s	1197 bits/s
192.168.31.137	142.251.127.108	28	16 kb	21	76	36.84%	14	8 kb	14	8 kb	124.805010600	9.056910	7370 bits/s	6981 bits/s
192.168.31.137	20.184.175.7	25	15 kb	4	64	39.06%	14	10 kb	11	4 kb	9.162905200	21.654013	3855 bits/s	1573 bits/s
192.168.31.137	52.123.129.124	27	14 kb	18	77	35.06%	16	7 kb	11	7 kb	52.79365100	968.533395	61 bits/s	55 bits/s
192.168.31.137	20.190.177.21	10	14 kb	28	42	23.81%	6	11 kb	4	2 kb	192.513177900	110.055216	817 bits/s	169 bits/s
192.168.31.137	2.16.158.162	18	13 kb	48	50	36.00%	6	3 kb	12	10 kb	853.355917700	120.097655	108 bits/s	365 bits/s
192.168.31.137	150.171.27.11	8	12 kb	27	29	27.59%	4	9 kb	4	3 kb	192.072121500	65.580278	1125 bits/s	368 bits/s
192.168.31.137	13.107.213.254	21	12 kb	10	55	38.18%	8	9 kb	13	9 kb	20.606319500	90.950715	276 bits/s	797 bits/s
192.168.31.137	20.184.175.20	21	11 kb	44	56	37.50%	12	7 kb	9	4 kb	689.195602100	3.215152	17 kbps	10 kbps
192.168.31.137	48.192.143.121	7	11 kb	41	24	29.17%	4	8 kb	3	3 kb	674.380747900	1.232254	52 kbps	17 kbps
192.168.31.137	2.16.158.179	34	10 kb	5	90	37.78%	20	6 kb	14	4 kb	11.004662600	676.858922	76 bits/s	45 bits/s
192.168.31.137	108.141.37.120	17	9 kb	8	44	38.64%	7	3 kb	10	6 kb	17.813230500	180.753842	115 bits/s	269 bits/s
192.168.31.137	2.16.158.152	11	9 kb	47	37	29.73%	5	3 kb	6	5 kb	852.610748300	297.945525	82 bits/s	146 bits/s
192.168.31.137	20.190.181.3	7	8 kb	38	34	20.59%	4	6 kb	3	3 kb	669.763195500	1.192039	38 kbps	18 kbps
192.168.31.137	95.101.75.50	11	8 kb	49	165	6.67%	4	2 kb	7	6 kb	916.902604800	236.145038	69 bits/s	199 bits/s
192.168.31.137	4.213.25.242	40	8 kb	3	80	50.00%	20	3 kb	20	5 kb	8.089993500	1140.249202	21 bits/s	31 bits/s
192.168.31.137	150.171.22.17	7	7 kb	45	28	25.00%	3	3 kb	4	4 kb	690.180812300	0.777743	29 kbps	41 kbps
192.168.31.137	95.100.102.49	9	6 kb	14	24	37.50%	3	1 kb	6	5 kb	28.614010700	65.396007	133 bits/s	618 bits/s
192.168.31.137	2.22.113.209	9	6 kb	12	25	36.00%	3	833 bytes	6	5 kb	27.571978500	66.438114	100 bits/s	624 bits/s
192.168.31.137	13.107.213.63	11	5 kb	39	27	40.74%	5	880 bytes	4	4 kb	671.249746400	0.570392	12 kbps	61 kbps
192.168.31.137	52.182.143.214	11	5 kb	16	27	40.74%	6	3 kb	5	2 kb	43.269619200	1.292635	19 kbps	12 kbps
192.168.31.137	20.42.73.24	11	5 kb	42	28	39.29%	6	3 kb	5	2 kb	674.99789600	1.062521	22 kbps	15 kbps
192.168.31.137	72.145.35.144	7	4 kb	40	24	29.17%	4	3 kb	3	1 kb	672.065376300	0.814114	28 kbps	12 kbps
192.168.31.137	13.107.6.254	12	4 kb	9	30	40.00%	6	1 kb	6	3 kb	19.739951800	130.808970	74 bits/s	176 bits/s
192.168.31.137	20.42.72.131	11	4 kb	46	29	37.93%	6	3 kb	5	789 bytes	790.18482600	1.106015	21 kbps	5706 bits/s
192.168.31.137	135.232.92.137	9	3 kb	1	23	39.13%	5	1 kb	4	2 kb	7.168476500	1.086809	7603 bits/s	13 kbps
192.168.31.137	72.154.71.10	6	3 kb	43	20	30.00%	3	859 bytes	2	2 kb	675.599744200	92.719387	74 bits/s	163 bits/s
192.168.31.137	2.16.158.33	2	3 kb	51	939	0.21%	1	1 kb	1	1 kb	1151.300824500	1.086400	3345 bits/s	3345 bits/s
192.168.31.137	204.79.197.222	9	3 kb	11	22	40.91%	5	2 kb	4	705 bytes	21.563597100	120.430417	117 bits/s	44 bits/s

After that, I reviewed the DNS traffic to identify the domains associated with the observed network connections.

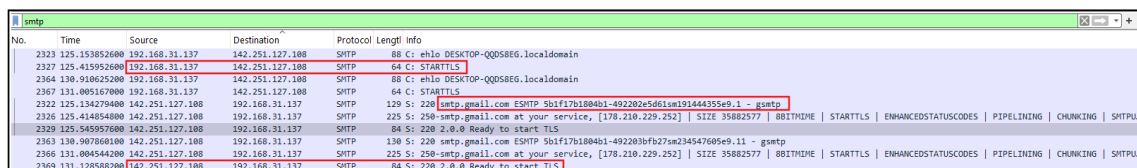
During the DNS analysis, I found multiple requests related to Microsoft services, Bing, MSN, Windows Update, and Akamai infrastructure. These results suggested that part of the observed traffic was related to normal operating system and cloud service activity.

Because most of the communication was encrypted, it was not possible to directly inspect the transmitted content from the packet capture alone.

To verify the exfiltration method used during the simulation, I searched for SMTP traffic using the following filter:

smtp

During the review, I identified SMTP communication between the host system and smtp.gmail.com. The traffic also showed the use of STARTTLS, which means the SMTP session was encrypted before data transmission occurred.

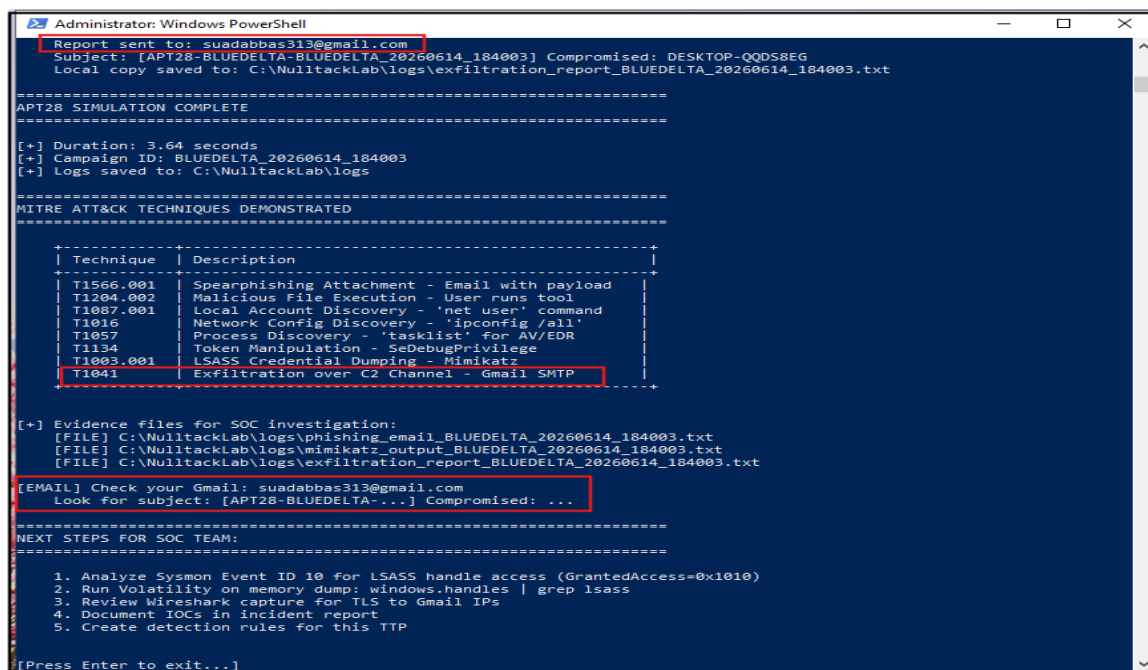


No.	Time	Source	Destination	Protocol	Length	Info
2323	125.153852600	192.168.31.137	142.251.127.108	SMTP	88	C: ehlo DESKTOP-QQ058EG.localdomain
2327	125.415952600	192.168.31.137	142.251.127.108	SMTP	64	C: STARTTLS
2364	130.910625200	192.168.31.137	142.251.127.108	SMTP	88	C: ehlo DESKTOP-QQ058EG.localdomain
2367	131.005167000	192.168.31.137	142.251.127.108	SMTP	64	C: STARTTLS
2362	125.134279400	142.251.127.108	192.168.31.137	SMTP	129	S: 220 smtp.gmail.com ESMTP Sblf17b1804b1-492202e5d61m191444355e9.1 - gsmt
2376	125.414054800	142.251.127.108	192.168.31.137	SMTP	225	S: 250-smtp.gmail.com at your service, [178.210.229.252] SIZE 35882577 8BITIME STARTTLS ENHANCEDSTATUSCODES PIPELINING CHUNKING SMTPU...
2329	125.545957600	142.251.127.108	192.168.31.137	SMTP	84	S: 220 2.0.0 Ready to start TLS
2363	130.907868100	142.251.127.108	192.168.31.137	SMTP	130	S: 220 smtp.gmail.com ESMTP Sblf17b1804b1-492203bfb27m234547605e9.11 - gsmt
2366	131.004544200	142.251.127.108	192.168.31.137	SMTP	225	S: 250-smtp.gmail.com at your service, [178.210.229.252] SIZE 35882577 8BITIME STARTTLS ENHANCEDSTATUSCODES PIPELINING CHUNKING SMTPU...
2369	131.128588200	142.251.127.108	192.168.31.137	SMTP	84	S: 220 2.0.0 Ready to start TLS

Figure 20 : While reviewing SMTP traffic, I identified communication with smtp.gmail.com. The session used STARTTLS, which showed that the connection was encrypted before the report was transmitted.

I then compared these findings with the output generated by the NulltackKatz simulation. The tool reported that the generated report was successfully sent to a Gmail account and identified the activity as **T1041 – Exfiltration Over C2 Channel**.

This provided additional confirmation that the simulated exfiltration activity was completed successfully during the exercise.



```

Administrator: Windows PowerShell
Report sent to: suadabbas313@gmail.com
Subject: [APT28-BLUEDELTA-BLUEDELTA_20260614_184003] Compromised: DESKTOP-QQ058EG
Local copy saved to: C:\NulltackLab\logs\exfiltration_report_BLUEDELTA_20260614_184003.txt

=====
APT28 SIMULATION COMPLETE
=====

[+] Duration: 3.64 seconds
[+] Campaign ID: BLUEDELTA_20260614_184003
[+] Logs saved to: C:\NulltackLab\logs

=====
MITRE ATT&CK TECHNIQUES DEMONSTRATED
=====

+-----+-----+
| Technique | Description |
+-----+-----+
| T1566.001 | Spearphishing Attachment - Email with payload |
| T1204.002 | Malicious File Execution - User runs tool |
| T1087.001 | Local Account Discovery - 'net user' command |
| T1016 | Network Config Discovery - 'ipconfig /all' |
| T1057 | Process Discovery - 'tasklist' for AV/EDR |
| T1134 | Token Manipulation - SeDebugPrivilege |
| T1003.001 | LSASS Credential Dumping - Mimikatz |
| T1041 | Exfiltration over C2 Channel - Gmail SMTP |
+-----+-----+

[+] Evidence files for SOC investigation:
[FILE] C:\NulltackLab\logs\phishing_email_BLUEDELTA_20260614_184003.txt
[FILE] C:\NulltackLab\logs\mimikatz_output_BLUEDELTA_20260614_184003.txt
[FILE] C:\NulltackLab\logs\exfiltration_report_BLUEDELTA_20260614_184003.txt

[EMAIL] Check your Gmail: suadabbas313@gmail.com
Look for subject: [APT28-BLUEDELTA-...] Compromised: ...

=====
NEXT STEPS FOR SOC TEAM:
=====

1. Analyze Sysmon Event ID 10 for LSASS handle access (GrantedAccess=0x1010)
2. Run Volatility on memory dump: windows.handles | grep lsass
3. Review Wireshark capture for TLS to Gmail IPs
4. Document IOCs in incident report
5. Create detection rules for this TTP

[Press Enter to exit...]

```

Figure 21 : After completing the simulation, I reviewed the NulltackKatz output and confirmed that the report was successfully sent to the configured Gmail account. The tool also identified the activity as T1041 (Exfiltration Over C2 Channel), which matched the SMTP and TLS traffic observed during the network investigation.

Based on the collected network evidence, the following answers were identified during the investigation.

Question	Answer
What destination IP range received the exfiltration?	SMTP communication was observed between the host system and 142.251.127.108 (smtp.gmail.com). Additional encrypted outbound connections were also identified during the investigation.
What destination port was used?	SMTP traffic was observed and the session used STARTTLS encryption.
Is the traffic encrypted?	Yes. TLSv1.2 and STARTTLS encrypted communications were identified during the packet capture.
How could a SOC detect this without decrypting?	A SOC could monitor SMTP connections, DNS requests, destination IP addresses, TLS metadata, and correlate network activity with host-based logs and process execution events.

Task 6 – Memory Forensics (Volatility 3)

Investigation Process

As part of the investigation, a memory image was acquired from the system using DumpIt. The memory capture was performed while Mimikatz was running in order to preserve any related artifacts in memory for analysis.

```

PS C:\Users\squad\Documents\Forensics> .\DumpIt.exe
DumpIt - v1.3.2.20110401 - One click memory memory dumper
Copyright (c) 2007 - 2011, Matthieu Suiche <http://www.msuiche.net>
Copyright (c) 2010 - 2011, MoonSols <http://www.moonsols.com>

Address space size:      2147483648 bytes (   2048 Mb)
Free space size:         4492251136 bytes (   4284 Mb)

* Destination = \\?\C:\Users\squad\Documents\Forensics\DESKTOP-QQDS8EG-20260618-012922.raw
--> Are you sure you want to continue? [y/n] y
+ Processing... Success.

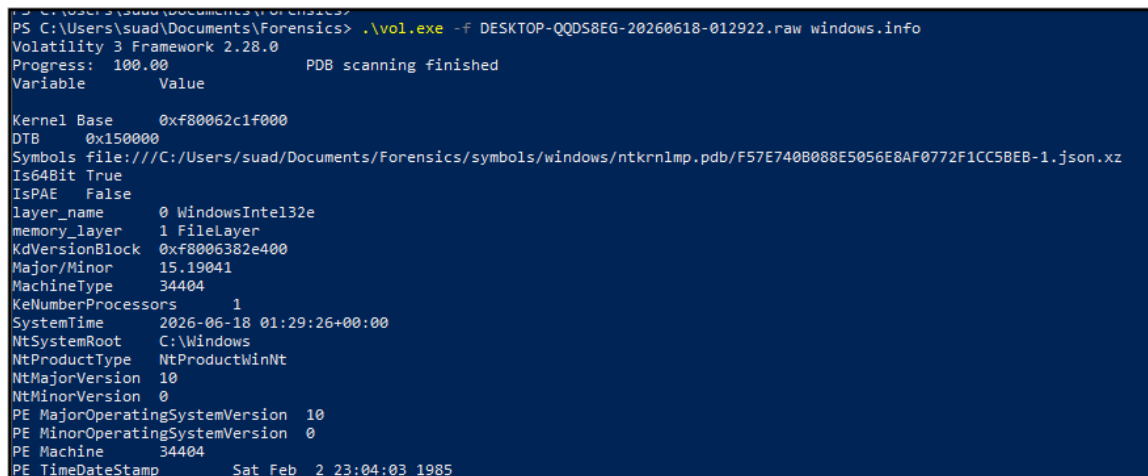
```

Figure 22 : Memory acquisition using DumpIt.

After the memory image was collected, Volatility 3 was used to analyze the memory dump. First, the operating system information was identified to verify that the memory image was successfully loaded and ready for analysis.

The following command was used:

```
.\vol.exe -f DESKTOP-QQDS8EG-20260618-012922.raw windows.info
```



```
PS C:\Users\squad\Documents\Forensics> .\vol.exe -f DESKTOP-QQDS8EG-20260618-012922.raw windows.info
Volatility 3 Framework 2.28.0
Progress: 100.00 PDB scanning finished
Variable Value
Kernel Base 0xf80062c1f000
DTB 0x150000
Symbols file:///C:/Users/squad/Documents/Forensics/symbols/windows/ntkrnlmp.pdb/F57E740B088E5056E8AF0772F1CC58EB-1.json.xz
Is64Bit True
IsPAE False
layer_name 0 WindowsIntel132e
memory_layer 1 FileLayer
KdVersionBlock 0xf8006382e400
Major/Minor 15.19041
MachineType 34404
KeNumberProcessors 1
SystemTime 2026-06-18 01:29:26+00:00
NtSystemRoot C:\Windows
NtProductType NtProductWinNt
NtMajorVersion 10
NtMinorVersion 0
PE MajorOperatingSystemVersion 10
PE MinorOperatingSystemVersion 0
PE Machine 34404
PE TimeDateStamp Sat Feb 2 23:04:03 1985
```

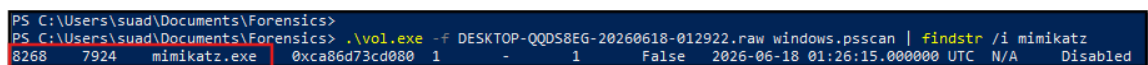
Figure 23 : Volatility successfully identifying Windows memory image information.

After confirming that the memory image was valid, process analysis was performed using the psscan plugin. This plugin was used to search memory for running and recently active processes.

The following command was used:

```
.\vol.exe -f DESKTOP-QQDS8EG-20260618-012922.raw windows.psscan | findstr /i mimikatz
```

The analysis identified an active Mimikatz process in memory. The process was detected as mimikatz.exe with PID 8268. This confirmed that the credential dumping tool was present in memory at the time the memory image was acquired.



```
PS C:\Users\squad\Documents\Forensics> .\vol.exe -f DESKTOP-QQDS8EG-20260618-012922.raw windows.psscan | findstr /i mimikatz
8268 7924 mimikatz.exe 0xca86d73cd080 1 - 1 False 2026-06-18 01:26:15.000000 UTC N/A Disabled
```

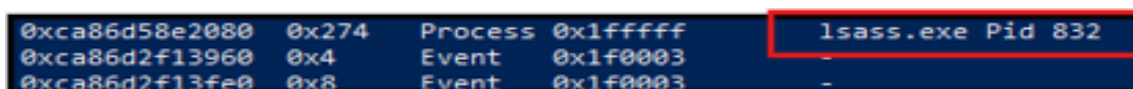
Figure 24 : Volatility psscan results showing mimikatz.exe in memory.

The memory analysis results were consistent with the evidence collected during previous investigation tasks. Sysmon logs confirmed Mimikatz execution, and the memory image provided additional evidence showing that the process was present in memory during acquisition.

After identifying the Mimikatz process in memory, additional analysis was performed using the Windows Handles plugin to review LSASS-related handles. The analysis confirmed that lsass.exe was present in memory and multiple handles associated with the process were identified. This provided additional evidence that the LSASS process was active at the time the memory capture was collected.

The following command was used:

```
.\vol.exe -f DESKTOP-QQDS8EG-20260618-012922.raw windows.handles | findstr /i lsass
```



Address	Offset	Process	Event
0xca86d58e2080	0x274	Process 0x1fffff	lsass.exe Pid 832
0xca86d2f13960	0x4	Event 0x1f0003	-
0xca86d2f13fe0	0x8	Event 0x1f0003	-

Figure 25 : Volatility handles analysis showing LSASS-related handles in memory.

Question	Answer
What is the PID of Mimikatz in memory?	8268
Does the memory dump show the executed commands?	The memory analysis confirmed the presence of mimikatz.exe in memory. However, no additional Mimikatz command-line evidence was identified during this stage of the investigation.

MITRE ATT&CK Mapping

Technique ID	Technique Name	Phase	Detection Method
T1566.001	Spearphishing Attachment	Initial Access	Email header review and phishing indicator analysis
T1204.002	User Execution	Execution	Evidence of user execution identified through simulation activity
T1003.001	OS Credential Dumping (LSASS)	Credential Access	Sysmon logs, Mimikatz execution evidence, Volatility memory analysis
T1041	Exfiltration Over C2 Channel	Exfiltration	TLS traffic review, SMTP analysis, Gmail communication, NulltackKatz exfiltration evidence

Detection Rule Writing

As part of the investigation, a Suricata rule was created to help detect encrypted connections to Gmail infrastructure that could potentially be used for data exfiltration. During the network analysis, SMTP and TLS communications with smtp.gmail.com were identified, therefore the following rule was written to generate an alert whenever a TLS connection is established to Gmail SMTP services.

```
alert tls $HOME_NET any -> $EXTERNAL_NET any (  
  msg:"Possible Gmail Exfiltration Activity";  
  tls.sni;  
  content:"smtp.gmail.com";  
  nocase;  
  sid:1000001;  
  rev:1;  
)
```


This rule can help security analysts identify systems that establish encrypted TLS connections to Gmail SMTP services. Such connections may be legitimate, but they should be reviewed when they occur during suspicious activity or security investigations.

Sysmon Detection Rule Checklist

Event ID	Detection Purpose
Event ID 1	Process Create — detect mimikatz.exe execution identified during the investigation.
Event ID 10	Process Access — monitor access attempts to lsass.exe memory used for credential dumping.
Event ID 3	Network Connect — monitor outbound SMTP and TLS communications with Gmail infrastructure.
Event ID 7	Image Load — detect suspicious DLL loading activity associated with malicious tools.

Task 7 APT28 Full Threat Intelligence Study

Part 1 – Group Identity & Attribution

Research Question	Answer
What are all known aliases for APT28 (Fancy Bear, BlueDelta, etc.)?	During the research, I found that APT28 is known by several names including Fancy Bear, Sofacy, Sednit, Pawn Storm, STRONTIUM, BlueDelta, and Forest Blizzard. Different security vendors use different names, but they all refer to the same threat group.
What is the exact military unit number and chain of command?	APT28 is widely attributed to the Russian General Staff Main Intelligence Directorate (GRU), specifically Unit 26165. The group is believed to operate under the Russian military intelligence structure.
What is the primary motivation and strategic objectives?	The primary objective of APT28 is cyber espionage. The group mainly targets government organizations, military entities, political institutions, media organizations, and critical infrastructure in order to collect intelligence and support Russian strategic interests.
What other GRU units operate alongside APT28?	Research showed that other GRU-linked units have been associated with cyber operations, including Unit 74455, which is commonly referred to as Sandworm Team.
What is the relationship with Sandworm Team (GRU Unit 74455)?	Both groups are linked to the GRU but perform different roles. APT28 is mainly associated with espionage and intelligence collection activities, while Sandworm is more frequently linked to disruptive and destructive cyber operations.

Part 2 – Historical Operations

Operation	Year	Target	Techniques Used
U.S. Presidential Election Interference	2016	DNC, Hillary Clinton campaign	Spearphishing emails, credential theft, information exfiltration
WADA / USADA Operations	2014-2018	Anti-doping agencies	Phishing campaigns, account compromise, data theft
OPCW / Spiez Laboratory Attacks	2018	Chemical weapons investigators	Credential harvesting, network reconnaissance, espionage activities
BlueDelta Credential Harvesting	2025	Turkish energy, European think tanks	Phishing emails, credential collection, targeted intelligence gathering
German Bundestag Cyber Espionage	2015	German Parliament	Spearphishing, malware deployment, long-term network access

Part 3: Technical TTP Analysis

As part of the threat intelligence research, I reviewed several MITRE ATT&CK techniques that have been associated with APT28 operations. The following table summarizes how these techniques have been used by the group and how some of them were also demonstrated during the BlueDelta simulation.

MITRE Technique ID	Technique Name	APT28 Specific Implementation
T1566.001	Spearphishing Attachment	APT28 commonly uses phishing emails with malicious attachments to trick victims into opening files and executing payloads. This was simulated in the exercise through the phishing email delivered to the victim.
T1204.001	Malicious Link	The group has been known to use malicious links that redirect victims to attacker-controlled content or credential harvesting pages.
T1003.001	LSASS Memory Dumping	APT28 has used credential dumping techniques to obtain account credentials from compromised systems. In this simulation, Mimikatz was used to access credential information from memory.
T1134	Token Manipulation	Threat actors may abuse access tokens and privileges to perform actions under another security context and maintain access to compromised systems.
T1041	Exfiltration over C2	APT28 has been observed collecting and exfiltrating stolen information from compromised systems. During the simulation, data was sent through Gmail SMTP traffic.
T1070.004	Redirect Victim to Legitimate Site	After phishing activity, victims may be redirected to legitimate websites to reduce suspicion and make the attack appear normal.

Part 4: Infrastructure & IoCs

As part of the threat intelligence research, I reviewed the Indicators of Compromise (IoCs) published by ESET in its *En Route with Sednit – Part 2* report. The report documents infrastructure and malware samples associated with Sednit (APT28), including domains, IP addresses, and malware hashes.

Research Item	Findings
List 5 domains used by APT28 (with purpose)	ciscohelpcenter.com – C2 domain microsoftsupp.com – C2 domain timezoneutc.com – C2 domain inteldrv64.com – C2 domain advpdxapi.com – C2 domain
List 5 IP addresses attributed to APT28	185.106.120.101 185.86.149.223 31.220.43.99 5.135.183.154 69.12.73.174
List 5 file hashes (MD5 or SHA256) from APT28 malware	072933fa35b585511003f36e3885563e 082141f1c24fb49981cc70a9ed50cda582ee0dd 08c4d755f14fd6f76c86da6eab1b5574dfbafd 0f04dad5194f97bb4f180d8f19196b04baee1b8 3403519fa3deedd07fb4c05d422a9f8c026cedbf

Part 5: Defensive Recommendations

Three specific detection rules to identify APT28 activity

- **Monitor Sysmon Event ID 10** to detect suspicious access to **lsass.exe**, which may indicate credential dumping activity associated with T1003.001.
- **Monitor Sysmon Event ID 1** for the execution of tools such as **mimikatz.exe**, PowerShell abuse, or suspicious processes launched from user directories.
- **Monitor outbound SMTP and TLS connections** to external destinations and investigate unusual encrypted communications that may indicate data exfiltration activity similar to T1041.

Three architectural changes to prevent APT28-style attacks

- **Implement Multi-Factor Authentication (MFA)** for email accounts and critical systems to reduce the risk of credential theft through phishing attacks.
- **Apply network segmentation** to limit lateral movement and restrict access between user workstations and sensitive systems.
- **Deploy centralized logging and monitoring solutions** such as Sysmon, SIEM, and network monitoring tools to improve detection and incident response capabilities.

Knowledge Graph (KG)

A Knowledge Graph was created to show the relationship between the threat actor, attack techniques, tools, and investigation findings identified during the scenario.

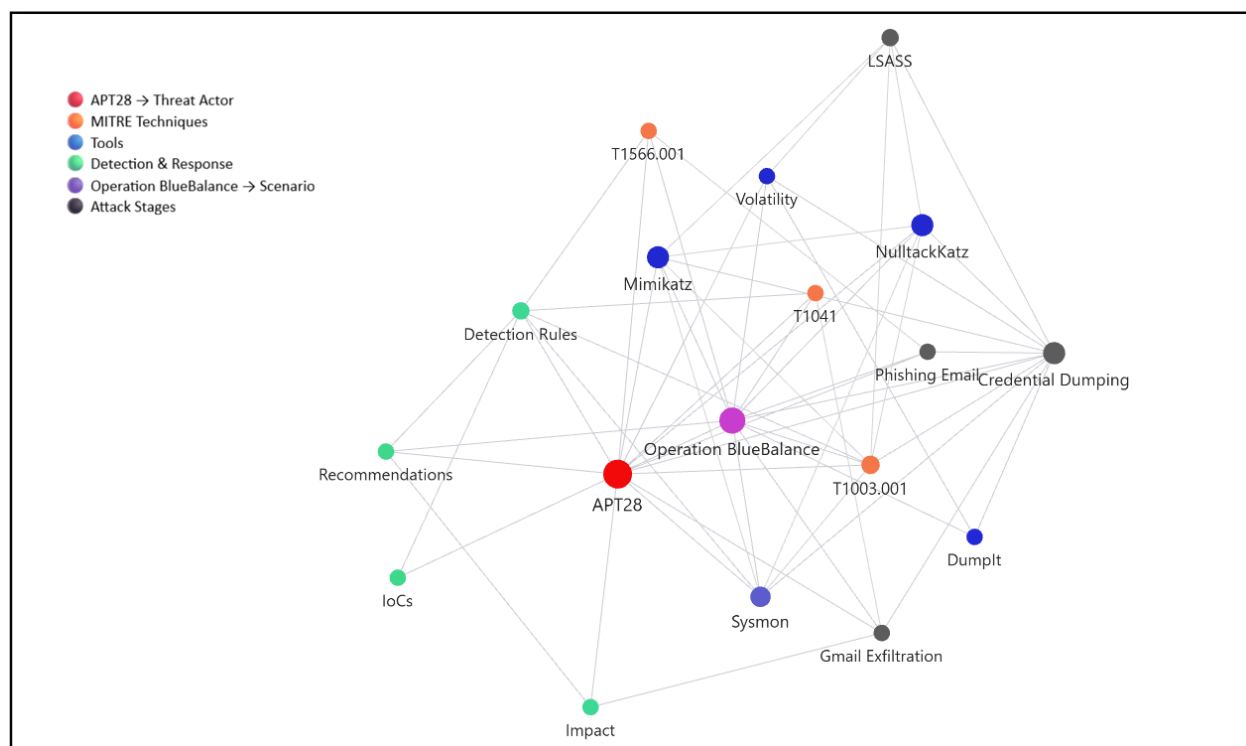


Figure 26 : Knowledge Graph showing relationships between APT28, MITRE ATT&CK techniques, tools, detection artifacts, and investigation findings.

Conclusion

During this investigation, I analyzed a simulated APT28 attack using the NulltackKatz tool. The attack started with a phishing email that convinced the victim to run a malicious file. After execution, the tool performed credential dumping activities and sent the collected information through an encrypted connection.

Throughout the investigation, I reviewed email evidence, Sysmon logs, network traffic, and memory artifacts to understand how the attack was performed. I also mapped the observed activities to the MITRE ATT&CK framework to identify the techniques used during each stage of the attack.

The Knowledge Graph was created to visualize the relationship between the attack stages, collected evidence, MITRE ATT&CK techniques, Indicators of Compromise (IoCs), and the potential impact of the attack. This helped provide a clearer understanding of the attack flow and how different findings were connected during the investigation.

Although Sysmon Event ID 10 was not generated during this exercise, other evidence sources including Sysmon process creation logs, Wireshark captures, and memory analysis provided sufficient visibility into the simulated attack. The investigation demonstrated how phishing, credential dumping, and exfiltration activities can be correlated and mapped to the MITRE ATT&CK framework.